

Machine Learning Based Trust Evaluation for Secure Cognitive Radio Sensor Networks with DDoS Detection

*Report submitted to the SASTRA Deemed to be University
in partial fulfillment of the requirements
for the award of the degree of*

B.Tech. Electronics & Communication Engineering

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May 2026



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Declaration

We declare that the project report titled “**Machine Learning Based Trust Evaluation for Secure Cognitive Radio Sensor Networks with DDoS Detection**” submitted by us is an original work done by us under the guidance of **Ms. M. Suba, Assistant Professor, Department of Electronics and Communication Engineering, Srinivasa Ramanujan Centre, SASTRA Deemed to be University, Kumbakonam** during the final semester of the academic year 2025–26. The work is original and wherever we have used materials from other sources, we have given due credit and cited them in the text of the report. This report has not formed the basis for the award of any degree, diploma, associate-ship, fellowship or other similar title to any candidate of any University.

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ACKNOWLEDGEMENT

We would like to express our sincere gratitude to our Honorable Chancellor, **Prof. R. Sethuraman**, for providing us with the opportunity and the necessary infrastructure to carry out this project as part of our curriculum.

We are deeply thankful to our Honorable Vice-Chancellor, **Dr. S. Vaidhyasubramaniam**, and **Dr. S. Swaminathan**, Dean, Planning & Development, for their constant encouragement and strategic support throughout our college journey. We also extend our sincere thanks to **Dr. R. Chandramouli**, Registrar, SASTRA Deemed to be University, for providing us with the opportunity to pursue this project.

Our heartfelt thanks go to **Dr. B. Santhi**, Dean, **Dr. A. Alli Rani**, Associate Dean and **Dr. D. Narasimhan**, Associate Dean, Srinivasa Ramanujan Centre, SASTRA Deemed to be University, for their encouragement and support during our project work.

Our guide, **Mrs. M. Suba**, Assistant Professor-II, Department of Electronics and Communication Engineering, Srinivasa Ramanujan Centre, was the driving force behind this project from the very beginning. Her deep insights and invaluable suggestions guided us in making steady progress throughout the project. We also thank the project review panel members for their valuable feedback and insights that significantly enhanced the quality of our work.

We would like to express our sincere gratitude to **Dr. K. Revathy**, Assistant Professor-II, Project Coordinator, Department of Electronics and Communication Engineering, Srinivasa Ramanujan Centre, for her invaluable guidance, continuous support, and encouragement towards the successful completion of our project report.

Finally, we gratefully acknowledge the unwavering encouragement and contributions from our families and friends, which played a vital role in the successful completion of this project.

We thank you all for giving us the opportunity to showcase our skills through this project.

Abstract

The growing proliferation of connected wireless devices and the resulting scarcity of radio spectrum resources have driven significant interest in Cognitive Radio Sensor Networks (CRSNs) as a means of enabling adaptive, opportunistic spectrum access alongside distributed environmental sensing. However, the open and decentralised architecture of these networks makes them highly susceptible to a range of security threats, including Distributed Denial of Service (DDoS) flooding, blackhole attacks, selective forwarding, and ON–OFF attacks. Conventional intrusion detection approaches designed for fixed wireless networks are ill-suited to the dynamic, resource-constrained CRSN environment. This project addresses the gap by proposing and implementing a machine learning based trust evaluation framework specifically designed for CRSN security enhancement.

The system simulates a network of approximately eighty sensor nodes, each assigned a behavioural class drawn from five categories: normal, blackhole, selective forwarding, ON–OFF attack, and DDoS flooding. Six behavioural features are extracted per node, namely forwarding ratio, drop ratio, communication delay, residual energy, previous trust score, and packets generated per slot. These features capture both packet-dropping and traffic-flooding behaviours within a single unified representation. Nine machine learning classifiers are benchmarked under stratified train-test evaluation, and the three highest-performing models are further refined through GridSearchCV hyperparameter optimisation. The tuned Gradient Boosting and Random Forest classifiers achieve classification accuracy of 95.83% with perfect precision (1.000) and AUC-ROC values of 0.969 and 0.948 respectively. Trust values are computed by temporally averaging per-node forwarding behaviour across ten observation windows, providing robustness against ON–OFF attackers. Trust-based cluster head selection ensures that data aggregation is always managed by the most reliable node in the network, substantially mitigating the influence of malicious nodes on routing decisions.

Specific Contribution:

- Designed and implemented the complete CRSN simulation environment, attack modelling, and six-feature extraction pipeline (Madhava Prasad K G).
- Benchmarked all nine ML classifiers, conducted GridSearchCV tuning, and produced comparative performance analysis (Raghul Raj B).
- Implemented temporal trust computation, cluster head selection, and result visualisation dashboards (Selvabharathi S).

Specific Learning:

- Gained practical experience in simulation-based dataset generation, feature engineering, and the application of ensemble machine learning to network security problems.

- Developed proficiency in hyperparameter optimisation using GridSearchCV and in interpreting AUC-ROC curves for imbalanced classification tasks.
- Understood the design principles of trust-based routing protocols and the challenges of detecting intermittent behavioural attack patterns in wireless networks.

Keywords: *Cognitive Radio Sensor Networks, Machine Learning, Trust Evaluation, DDoS Detection, Secure Clustering, Random Forest, Gradient Boosting, Malicious Node Detection.*

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CHAPTER 1

INTRODUCTION

1.1 BACKGROUND AND MOTIVATION

Modern wireless networks face increasing pressure due to the high number of devices competing for limited radio spectrum. Smartphones, sensors, vehicle communication systems, smart meters, healthcare devices, and countless IoT devices all require dependable, low-latency wireless connections. However, the available spectrum for unlicensed and secondary users is highly limited. Traditional fixed spectrum allocation schemes, which dedicate specific frequency bands to licensed primary users regardless of usage, are inefficient. Urban and suburban studies show large parts of licensed spectrum sit unused much of the time, even as unlicensed bands become congested, causing performance issues for legitimate users.

This paradox of scarcity and underuse has driven the development of dynamic spectrum access (DSA) technologies, especially Cognitive Radio (CR). Formally introduced by Mitola and Maguire, and further developed by Haykin, CR is a radio tech that senses its electromagnetic environment, learns from it, and adapts its transmission parameters intelligently. It exploits spectral opportunities while safeguarding primary users. By sensing, reasoning, and acting on spectral availability, CR transforms the radio channel from a fixed resource into a shared, flexible medium that supports many secondary users beyond traditional schemes.

This project is motivated by the observation that security issues in CR-based sensor networks (CRSNs) have not been fully addressed. While trust-based frameworks and machine learning methods have been studied separately, their integration into a comprehensive system that tackles the full range of CRSN security threats—including stealthy DDoS flooding attacks—is limited. This work proposes a complete ML-driven trust evaluation framework that models five different behavioural classes, extracts six key features, compares nine classifiers, and combines temporal trust assessment with cluster head selection.

1.2 SPECTRUM SCARCITY AND COGNITIVE RADIO

The ITU reports that demand for wireless spectrum will increase dramatically over the next decade, mainly due to expanding broadband mobile services and IoT connectivity. The traditional fixed allocation model, where regulatory agencies assign specific frequencies to particular services indefinitely, cannot keep up with this growth without either new spectrum allocations—which are physically limited—or a shift to dynamic sharing. Cognitive radio offers this change by allowing secondary users to transmit opportunistically in frequency bands that are momentarily free from primary users, vacating immediately when primary users reappear.

A cognitive radio system operates through a continuous cycle of sensing, learning, and adapting. Sensing involves monitoring the environment's power spectral density to detect

primary signals. Learning creates a model of how spectrum usage varies over time and space, helping predict when and where opportunities will arise. Adapting then adjusts transmission parameters—such as frequency, power, modulation, and coding—to exploit these opportunities while avoiding interference. According to Akyildiz et al. [6], future cognitive radio networks will likely operate across multiple frequency bands simultaneously, requiring advanced coordination methods that are themselves susceptible to malicious attacks.

1.3 CRSNS IN THE IOT LANDSCAPE

Cognitive Radio Sensor Networks (Crsns) combine cognitive radio technology with wireless sensor network (WSN) architectures. Unlike traditional WSNS, where sensor nodes collect environmental data and send it via multi-hop routing to a sink or gateway on fixed frequencies that may be jammed, congested, or crowded by other wireless systems, CRSNs allow sensor nodes to dynamically select transmission frequencies through real-time spectrum sensing. This capability significantly enhances spectral efficiency and resistance to interference.

CRSNS are suited for a wide array of critical applications. In smart city infrastructure, they monitor parameters such as air quality, traffic, and noise in densely populated areas with congested spectrum. In industrial IoT settings, machinery equipped with CRSNS oversee production parameters on factory floors with many co-existing wireless systems. In agriculture, CRSN nodes track soil moisture, temperature, and crop health across large rural regions, often shared with primary users like meteorological agencies or broadcasters. Healthcare applications involve body-area CRSNS that transmit vital signs while avoiding interference with hospital communication systems. During disasters, CRSN nodes provide situational awareness in environments where traditional networks are compromised, utilising available spectrum opportunistically.

As CRSNs become more integrated into the IoT landscape, their usefulness increases, but so does their vulnerability. Growing numbers of interconnected nodes embedded in critical infrastructure pose significant security risks. For instance, a hacked CRSN node in a smart grid could disable alarms; a DDoS attack on healthcare CRSNS might delay vital patient data; and a blackhole attack in disaster zones could prevent rescue teams from receiving crucial information.

1.4 SECURITY CHALLENGES IN CRSNS

The security challenges specific to CRSNs arise from the combination of several factors that do not occur simultaneously in conventional networks. First, the open wireless channel is inherently accessible to all parties within radio range, making it easy for adversaries to monitor traffic, inject signals, or disrupt communication without physical access to the network. Second, the dynamic spectrum access mechanism itself is a target: primary user emulation (PUE) attacks involve malicious nodes impersonating primary users to force legitimate secondary users to vacate spectral bands, effectively conducting a denial-of-service attack through the spectrum sensing mechanism. Third, sensor nodes are typically resource-constrained devices with limited computational power, memory, and battery

capacity, making it impractical to run the cryptographic protocols and intrusion detection systems routinely deployed in wired networks. Fourth, the absence of a centralised authority means that routing decisions are made cooperatively, creating opportunities for malicious nodes to manipulate routing tables or selectively drop packets with limited accountability.

Trust management has emerged as one of the most promising paradigms for addressing these challenges in resource-constrained distributed networks. The fundamental concept, drawing on the work discussed in [1] and [4], is that each node maintains a reputation model for its neighbours based on observed behaviour rather than relying exclusively on cryptographic credentials. Nodes that consistently forward packets as expected accumulate high trust scores, while nodes that exhibit anomalous behaviour—dropping packets, generating excessive traffic, or alternating between cooperative and malicious states—accumulate low trust scores. Routing protocols can then weight paths by the trust values of constituent nodes, effectively isolating malicious nodes from the data forwarding path without requiring centralised coordination.

1.5 DDOS AND OTHER ATTACK PATTERNS

This project focuses on four attack types that pose the most significant and varied security threats to CRSNs. Understanding each attack’s behavior is crucial for creating a detection system capable of reliably differentiating malicious activities from normal network variations.

Blackhole Attacks: A blackhole node accepts all incoming data packets from neighboring nodes but silently discards them instead of forwarding to the destination. To other nodes, it appears as a capable router, and routing algorithms that rely solely on link quality or hop count tend to route traffic through it. This creates a network region from which data cannot escape—a forwarding hole. In our simulation, blackhole nodes have a forwarding ratio of zero, which is a key indicator for detection. As noted in [1], detecting blackhole attacks in CRSNs is challenging because legitimate packet loss during spectrum handoff can temporarily lower forwarding ratios, making it necessary for trust systems to differentiate between brief losses and persistent ones.

Selective Forwarding Attacks: Instead of dropping all packets, these nodes selectively drop a subset—usually between 30% and 60%—of received traffic to evade simple detection mechanisms while still harming network performance. They may drop packets randomly or target specific nodes or content. In our simulation, these nodes have forwarding ratios from 0.4 to 0.7, placing them in a grey area between clearly malicious and normal behavior. Detecting such nodes requires a classification model that considers drop ratio, delay, and trust history to identify statistically unusual forwarding patterns.

ON–OFF Attacks: These sophisticated adversaries alternate between cooperative and malicious behaviors. During cooperative phases, they forward packets normally and earn high trust scores. During malicious phases, they aggressively drop packets. By timing their cooperation to coincide with trust evaluation periods, they can sustain high trust scores while causing damage during malicious episodes. Single-window trust evaluations are

vulnerable to this approach; hence, our framework averages forwarding behavior over ten consecutive windows before assigning trust. According to [4], the temporal granularity of trust observation is a critical factor for trust-based security systems.

DDoS Flooding Attacks: Unlike the other attack types, DDoS attacks inject large volumes of self-generated traffic, overwhelming bandwidth, processing power, and energy resources of neighboring nodes and the overall routing infrastructure. In our simulation, DDoS nodes generate between 40 and 60 packets per time slot, compared to 5 to 10 for normal nodes. Although DDoS nodes may maintain high forwarding ratios because they genuinely forward most packets, the volume of traffic they produce constitutes the attack. This makes them undetectable by systems that only monitor forwarding ratios and highlights the importance of including the generated packets as a distinctive feature for DDoS detection. The traffic analysis methods align with those discussed in [2] and [5].

1.6 PROPOSED SOLUTION OVERVIEW

1. This project proposes a unified machine learning trust evaluation framework to secure CRSNs, incorporating five key components: a simulation environment generating realistic behavioural data for nodes across five classes; a feature extraction module identifying six attributes per node; a benchmarking pipeline evaluating nine ML models; a hyperparameter optimisation layer refining top models with GridSearchCV; and a trust computation and cluster head selection module translating results into routing security decisions.
2. Building on the approach in [1], which showed trust-based clustering improves packet delivery under attacks, this framework adds an ML classification layer for DDoS detection based on packet generation features. Inspired by ensemble learning in [7], it identifies Random Forest and Gradient Boosting as most effective, justified through AUC-ROC analysis rather than accuracy.
3. The five main contributions are:
 1. A simulation framework modelling 80 nodes in five behavioural classes with realistic packet profiles.
 2. A six-feature extraction pipeline capturing behaviours like packet dropping and traffic flooding, plus contextual node states.
 3. A benchmarking study of nine classifiers using stratified splits and five metrics, including AUC-ROC.
 4. Hyperparameter tuning of the top three classifiers with Gridsearchcv to maximise F1-score, improving sensitivity to malicious nodes.
 5. A temporal trust evaluation scheme averaging node behaviour over ten windows, providing robustness against ON-OFF attacks, coupled with trust-based cluster head selection for secure routing.

CHAPTER 2

OBJECTIVES

The objectives of this project are formulated to address the specific security gaps identified in the introduction and to produce a system that is both technically rigorous and practically deployable in CRSN environments.

Primary Objectives:

1. Design and implement a simulation-based CRSN environment that accurately models packet forwarding and generation behaviours of nodes across five different behavioural classes, including four attack variants.
2. Develop a six-feature extraction method capturing both packet-dropping and traffic-flooding attack signatures within a unified representation, allowing a single ML model to detect all four attack types simultaneously.
3. Benchmark nine ML classifiers using stratified evaluation to identify the most effective models for CRSN malicious node detection, based on accuracy, F1-score, precision, recall, and AUC-ROC.
4. Optimise the best models through GridSearchCV hyperparameter tuning and demonstrate improvements in AUC-ROC and generalisation performance compared to untuned baselines.
5. Create a trust computation mechanism that averages node forwarding behaviour over multiple observation windows to increase robustness against ON–OFF attacks, and use these trust scores for secure cluster head selection.

Secondary Objectives:

6. Perform feature importance analysis for the top three classifiers, validating each feature's contribution, especially the role of generated packets in DDoS detection.
7. Analyse the confusion matrix of the best model to distinguish false positives from false negatives and interpret their security implications for CRSN routing.
8. Provide a complete, open-source implementation using Python libraries (NumPy, pandas, scikit-learn, matplotlib) with fixed seeds to facilitate future reproduction and extension of the results.

CHAPTER 3

LITERATURE SURVEY

Research in Cognitive Radio Sensor Network security has evolved along two largely parallel tracks: trust-based management approaches derived from wireless sensor network security literature, and machine learning-based anomaly detection approaches adapted from network intrusion detection research. This chapter surveys the most relevant contributions from both tracks, identifies the gaps that the proposed framework addresses, and provides the foundation for the design decisions made in Chapter 4.

3.1 TRUST-BASED SECURITY IN SENSOR NETWORKS

The concept of trust management in sensor networks was formalised to address the limitations of purely cryptographic security approaches, which require key infrastructure and extensive computational resources that are often impractical for resource- constrained sensor nodes. Instead, trust systems develop a reputation model for each node based on observed behaviours. This approach allows the network to identify and isolate misbehaving nodes without relying on a central authority.

The security- enhanced clustering scheme for routing in cognitive radio sensor networks discussed in [1] is one of the most relevant contributions to this project. The authors propose a trust model combining direct trust observations—based on packet forwarding behaviour monitored over successive time windows—with indirect trust from recommendations by shared neighbours. This scheme shows improved packet delivery ratios and reduced end- to- end delays despite blackhole and selective forwarding attacks. It also introduces trust- weighted cluster head selection to ensure the most reliable node manages data aggregation. This work directly influences the trust computation and cluster head selection mechanisms in Chapters 4. 5 and 4. 6 of this project.

The trust- based secure routing framework in [2] extends the analysis to environments with mixed attack types, where different nodes employ various strategies simultaneously. The authors demonstrate that adaptive trust decay—where trust scores decrease over time unless updated by recent positive observations—significantly enhances detection of ON–OFF attacks compared to cumulative trust models that weigh all past observations equally. This concept has influenced our temporal averaging method, which gives roughly equal weight to the most recent forwarding behaviours and previous observations within a ten- window period.

Kranthikumar and Velusamy developed a trust- aware, energy- efficient fuzzy clustering protocol that incorporates both trust and residual energy into the cluster head election process. Their findings show that optimising solely for trust can lead to selecting cluster heads with low remaining energy, causing frequent re- clustering and additional overhead. This dual- objective approach motivates the inclusion of residual energy as a key feature in our extraction process, even though our primary goal is detection rather than routing optimisation.

3.2 MACHINE LEARNING FOR NETWORK ANOMALY DETECTION

The use of machine learning in network intrusion detection has a long-standing presence in security literature. However, adapting it to the unique traits of CRSNs—such as small node counts, resource limitations, dynamic topologies, and overlapping attack types—is relatively recent. As noted in [6], future CRSN security systems must handle complex, multi-dimensional attack patterns that go beyond simple threshold-based detection.

Ensemble learning methods, like those in [7], show that combining classifiers via voting or stacking consistently outperforms individual models in network security tasks. The Random Forest model, in particular, performs well across various intrusion detection benchmarks due to its ability to utilise feature subsets through random subspace sampling. This makes it robust against feature correlations common in CRSN behavioural data—where forward ratio and drop ratio are mathematically related (sum to one) but each provides unique information for different attack types.

The approach in [2], which benchmarks SVM, Decision Tree, and ensemble classifiers on a synthetic CRSN dataset with three attack types, reveals that no single classifier dominates. Model choice should be based on the AUC-ROC metric rather than accuracy alone, especially when attack classes are sparse. This insight led to the nine-classifier benchmarking methodology and the AUC-ROC-focused model selection used in Chapter 5.

Research on DDoS detection, such as in [5], indicates that traffic volume features—specifically packets per unit time—are the most effective indicators of flooding attacks in sensor networks. Systems that rely only on forwarding ratio miss DDoS nodes because these nodes can forward normal traffic rates while still being malicious. This crucial observation motivated the inclusion of packets generated as a dedicated feature in this project's data extraction process.

3.3 COGNITIVE RADIO FOUNDATIONS

The cognitive radio paradigm is based on foundational work by Haykin [8], who defined cognitive radio as a wireless communication system empowered by a 'brain' capable of awareness, learning, and adaptation. Mitola and Maguire [9] initially envisioned software-defined radio systems that could personalise their operation by sensing and adapting to user context. Akyildiz et al. [6] expanded this concept into the multi-hop networking arena, describing cognitive radio mesh network architecture and highlighting four key technical functions- spectral sensing, management, mobility, and sharing- that set CR networks apart from traditional wireless systems. Specific challenges of spectrum sensing in secondary user environments are discussed in [11] and [12], which establish energy detection as the primary sensing technique and point out its vulnerability to primary user emulation attacks.

3.4 SUMMARY AND RESEARCH GAP

Table 3.1 summarises the key contributions from the literature survey and identifies the specific limitations that the proposed framework addresses.

Table 3.1 Summary of Related Literature

Reference	Key Contribution	Attack Types Addressed	Limitation
[1]	Trust-weighted cluster head selection in CRSN	Blackhole, Selective	No ML; no DDoS detection
[2]	Adaptive trust decay for ON–OFF detection	Blackhole, ON–OFF	No ML; limited feature set
[5]	Traffic volume DDoS detection in WSN	DDoS	No trust integration
[6]	Cognitive radio network survey	General	No security focus
[7]	Ensemble ML for intrusion detection	General	Not specific to CRSN
This work	ML + trust evaluation; 9 classifiers; 6 features; 4 attack types	All four types	Simulation scale (80 nodes)

CHAPTER 4

METHODOLOGY

The proposed framework is implemented as a seven-stage computational pipeline, progressing from raw network simulation through feature extraction, ML training and evaluation, hyperparameter optimisation, trust computation, cluster head selection, and result visualisation. Each stage is described in detail in the following sections.

4.1 NETWORK SIMULATION AND DATASET GENERATION

The simulation models a Cognitive Radio Sensor Network with around 80 sensor nodes randomly distributed in a 50×50 unit area. Node positions are independently sampled from a uniform distribution, mimicking the typical unplanned deployment in environmental monitoring and disaster response. Implemented in Python with Numpy, the simulation uses a fixed seed (`np.random.seed(42)`) to guarantee reproducibility.

Each node is assigned a behaviour class based on a discrete probability distribution that mirrors realistic CRSN security scenarios: 50% normal, 15% blackhole, 15% selective forwarding, 10% ON–OFF attack, and 10% Ddos. This setup ensures the majority of nodes are normal, aligning with real-world network assumptions, while also providing enough attack samples to effectively train classifiers.

Table 4.1 Attack Type Behaviour Parameters

Attack Type	Class Weight	Forwarding Ratio	Packets Generated / slot
Normal	50%	0.85 – 1.00	5 – 10
Blackhole	15%	≈ 0.00	5 – 10
Selective Forwarding	15%	0.40 – 0.70	5 – 10
ON–OFF Attack	10%	0.30 – 1.00 (variable)	5 – 10
DDoS Flooding	10%	0.80 – 1.00	40 – 60

For each node, the number of packets sent per time window is uniformly sampled from the range [80, 120], representing typical CRSN traffic loads. The packets forwarded are determined based on the per-class forwarding ratio profile outlined in Table 4.1. DDoS nodes also generate a large amount of self-originated traffic, indicated by the packets generated attribute, which sets them apart from other node types that produce only 5 to 10 packets per slot. This behavioural difference forms the basis for the DDoS detection capability of the system, inspired by the traffic-volume analysis methods discussed in [5].

4.2 FEATURE EXTRACTION

Six behavioural features are extracted from each node using raw packet statistics. The selection of features follows three principles: each should represent a unique aspect of node behaviour, together they must distinguish all four attack types effectively, and they should be derivable from local measurements without needing global network information.

Table 4.2 Dataset Feature Description

Feature	Formula / Range	Primary Role	Most Informative For
Forward Ratio	Forwarded / Sent; [0, 1]	Primary forwarding metric	Blackhole, Selective Forwarding
Drop Ratio	1 – Forward Ratio; [0, 1]	Inverse forwarding metric	Blackhole, Selective Forwarding
Delay	Uniform [0.1, 1.5] s	Congestion indicator	DDoS, ON–OFF
Residual Energy	Uniform [0.2, 1.0]	Resource depletion monitor	DDoS
Previous Trust	Uniform [0.3, 0.9]	Behavioural memory	ON–OFF
Packets Generated	Uniform [5–10] or [40–60]	Traffic injection measure	DDoS (primary discriminator)

Although the forward ratio and drop ratio are mathematically complementary (forward ratio + drop ratio = 1.0), both are kept in the feature set. Tree-based ensemble models can leverage each feature independently via different split points, creating redundant discriminating paths that enhance classification robustness. This approach aligns with the method in [7], where correlated features were kept if permutation importance analysis verified their individual contribution after conditioning on other features.

4.3 MACHINE LEARNING MODEL TRAINING

The feature matrix X (80 rows $\times$ 6 columns) and binary target vector y (where 0 indicates normal and 1 indicates malicious, with all four attack types grouped as malicious) are split into a training set (70%) and a test set (30%) using stratified sampling. This stratification maintains the 50:50 class ratio of the entire dataset in both splits, preventing label imbalance artifacts that could artificially inflate accuracy for the majority class. Randomisation is seeded at 42 for consistency.

Nine classifiers from scikit-learn are trained and evaluated: Logistic Regression (max_iter=1000), K-Nearest Neighbors (k=5), Support Vector Machine (RBF kernel), Decision Tree, Random Forest (150 estimators), Gradient Boosting, AdaBoost, Gaussian Naive Bayes, and Multi-Layer Perceptron (max_iter=1000). These classifiers cover linear models, kernel methods, single-tree models, ensemble methods, probabilistic models, and neural networks, providing a comprehensive overview of machine learning options and supporting informed model selection.

All models are tested on the 30% hold-out set using five metrics: accuracy, F1-score, precision, recall, and AUC-ROC. As discussed in [2], accuracy alone is insufficient for model comparison on small test sets, where high feature separability can lead multiple classifiers to similar accuracy scores despite notable differences in probabilistic calibration and generalisation. Therefore, AUC-ROC is used as the main criterion for model selection in this study.

4.4 HYPERPARAMETER OPTIMISATION

The top three classifiers—Decision Tree, Random Forest, and Gradient Boosting—are further optimised using GridSearchCV with 5-fold stratified cross-validation, focusing on the F1-score as the criterion. The F1-score was selected over accuracy because it balances precision and recall, ensuring the model maintains detection sensitivity for the malicious class without sacrificing specificity.

Decision Tree: The grid search considers $\text{max_depth} \in \{5, 10, 15, 20\}$, $\text{min_samples_split} \in \{2, 5, 10\}$, and $\text{min_samples_leaf} \in \{1, 2, 4\}$. The best settings ($\text{max_depth}=5$, $\text{min_samples_split}=10$, $\text{min_samples_leaf}=2$) limit the depth to avoid overfitting on the 56-sample training set, while still capturing all four attack patterns.

Random Forest: The search space also includes $\text{n_estimators} \in \{50, 100, 150, 200\}$. The optimal parameters ($\text{n_estimators}=50$, $\text{max_depth}=5$) match the accuracy of the baseline with 150 estimators but cut inference time by around two-thirds, indicating ensemble diversity saturates below the default size for this dataset.

Gradient Boosting: This expands the search to $\text{learning_rate} \in \{0.01, 0.05, 0.10\}$. The best configuration ($\text{n_estimators}=50$, $\text{learning_rate}=0.01$, $\text{max_depth}=3$) uses a conservative boosting approach, preventing overfitting on the small dataset through slow, incremental improvements.

4.5 TRUST COMPUTATION AND NODE CLASSIFICATION

The trust computation module functions independently from the ML classification pipeline, relying on raw forwarding ratio observations rather than classification labels. This separation ensures the trust system stays operational even without a trained ML model, allowing the outputs to be compared or combined as needed in the deployment.

For each of ten consecutive time windows, the forwarding ratio of each node is measured and slightly perturbed by a multiplicative noise factor sampled uniformly from $[0.8, 1.0]$. This noise models natural variability in packet delivery caused by factors such as channel fading, spectrum handoff, or transient congestion, rather than malicious activity. The final trust score for each node is calculated as the average forwarding ratio over all ten windows. This temporal averaging enhances robustness against ON–OFF attackers, as explained in [4], since their cooperative periods where they behave normally are not enough to elevate the ten-window average above the detection threshold unless they cooperate most of the time.

Nodes with mean trust scores below the threshold $\theta = 0.5$ are labelled malicious and excluded from routing and data aggregation candidate pools. The 0.5 threshold was chosen to position the classification boundary midway between the trust score range of normal nodes (roughly 0.7 to 0.9 after averaging) and that of blackhole or selective forwarding nodes (roughly 0.0 to 0.4 after averaging), providing balanced separation margins.

4.6 CLUSTER HEAD SELECTION

Nodes classified as trustworthy by the trust threshold select a cluster head based on the highest mean trust score. This greedy trust-maximisation approach ensures the data aggregation function is assigned to the node with the strongest history of cooperative forwarding, reducing the likelihood of the cluster head selectively dropping data.

The trust-based cluster head selection in this project aligns with principles in [1] and [3], where trust-weighted criteria notably improve packet delivery ratios over energy-only or random methods. Although this project doesn't implement the full multi-cluster CRSN routing protocol, it demonstrates how trust evaluation can enhance routing security through cluster head selection.

The overall workflow, from raw data to trust-based cluster head selection, is summarised in Fig. 4.1. The system design is modular, enabling future researchers to swap out feature extraction, classifiers, or trust functions independently, without redesigning the entire framework.

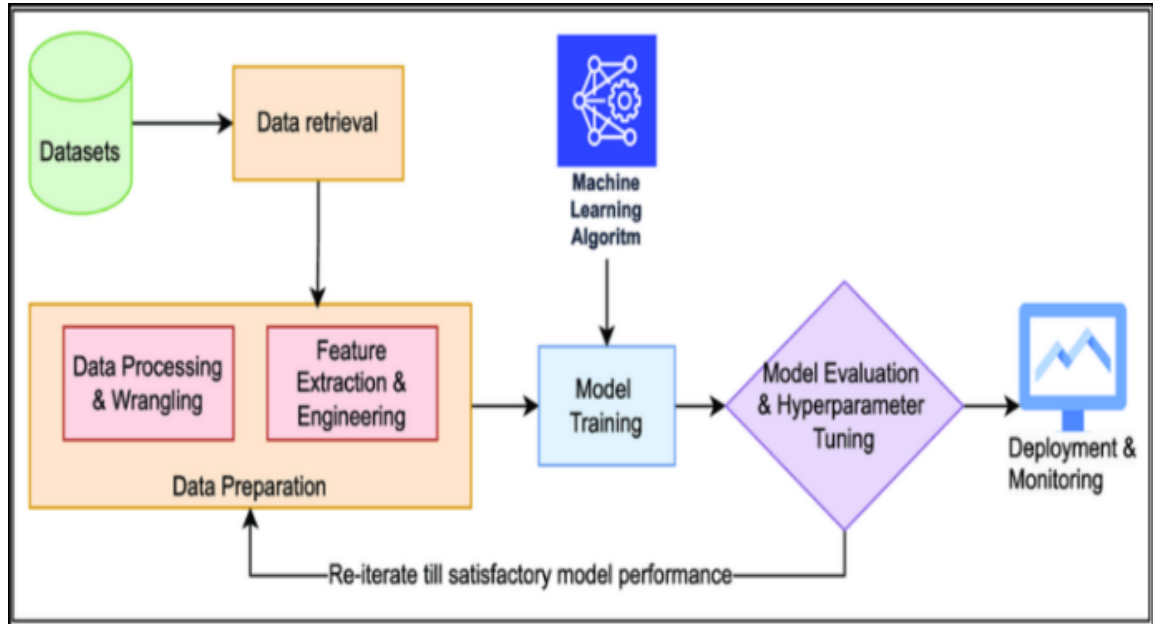


Fig. 4.1 Proposed System Workflow: Data Generation → Feature Extraction → ML Classification → Trust Computation → Cluster Head Selection

CHAPTER 5

RESULTS AND DISCUSSION

This chapter presents and interprets the experimental results of the proposed ML-based trust evaluation framework. All results are computed on the 30% stratified test set (24 samples) and are fully reproducible under the fixed random seed conditions described in Chapter 4. The discussion addresses baseline model performance, hyperparameter tuning outcomes, feature importance analysis, confusion matrix characteristics, trust distribution behaviour, and a comprehensive comparative analysis that culminates in the identification and justification of the best-performing model.

5.1 BASELINE MODEL PERFORMANCE

The classification accuracy of all nine baseline models is shown in Fig. 5.1 and summarised in Table 5.1. The results reveal a clear performance stratification across model families. Ensemble tree-based models—Decision Tree, Random Forest, and Gradient Boosting—achieve identical accuracy of 95.83%, correctly classifying 23 of 24 test samples. Linear and probabilistic models—Logistic Regression, AdaBoost, and Gaussian Naive Bayes—attain 91.67% accuracy. The Neural Network achieves 87.50%, reflecting underfitting on the 56-sample training set. KNN reaches 83.33%, suggesting sensitivity to the feature correlation between forward ratio and drop ratio. Support Vector Machine achieves only 54.17%, indicating that the RBF kernel without explicit feature scaling is poorly suited to this feature space.

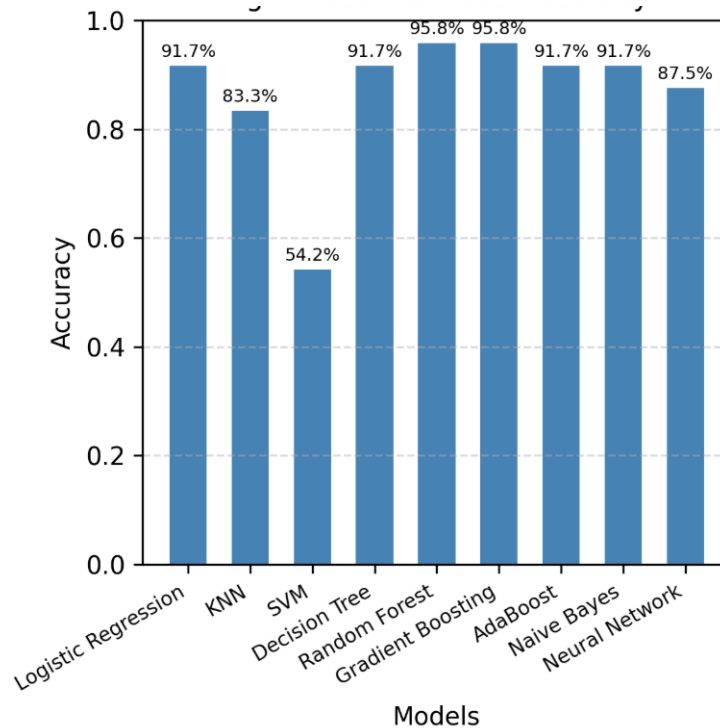


Fig. 5.1 Baseline Model Accuracy Comparison (All 9 Classifiers)

Table 5.1 Baseline Classification Accuracy of Nine ML Models

Classifier	Accuracy (%)	Notes
Logistic Regression	91.67	Linear decision boundary; misses nonlinear DDoS patterns
K-Nearest Neighbors (k=5)	83.33	Distance metric sensitive to correlated features
Support Vector Machine (RBF)	54.17	Requires normalisation; poorly calibrated without scaling
Decision Tree	95.83	Strong but overfit risk without depth constraint
Random Forest	95.83	Ensemble diversity improves generalisation
Gradient Boosting	95.83	Sequential refinement on hard samples
AdaBoost	91.67	Sensitive to noise in minority attack classes
Gaussian Naive Bayes	91.67	Independence assumption violated by feature correlation
Neural Network (MLP)	87.50	Underfits with only 56 training samples

The identical accuracy of Decision Tree, Random Forest, and Gradient Boosting reflects the strong separability of the forward ratio feature on this 24-sample test set: blackhole nodes cluster at forward ratio ≈ 0 while normal nodes cluster above 0.85, creating a decision boundary that all tree-based methods find trivially. This result motivates the use of AUC-ROC as the primary differentiating criterion in subsequent analysis, as argued in [2].

5.2 HYPERPARAMETER TUNING RESULTS

Following baseline evaluation, the three tied models were subjected to GridSearchCV optimisation as described in Chapter 4.4. Figure 5.2 compares the accuracy of the three tuned models, while Table 5.2 presents the comprehensive metric profiles of all six model variants (baseline and tuned versions of each top model).

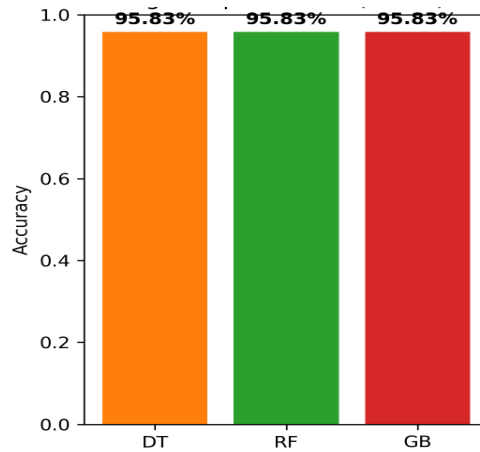


Fig. 5.2 Top 3 Tuned Model Comparison (DT, RF, GB)

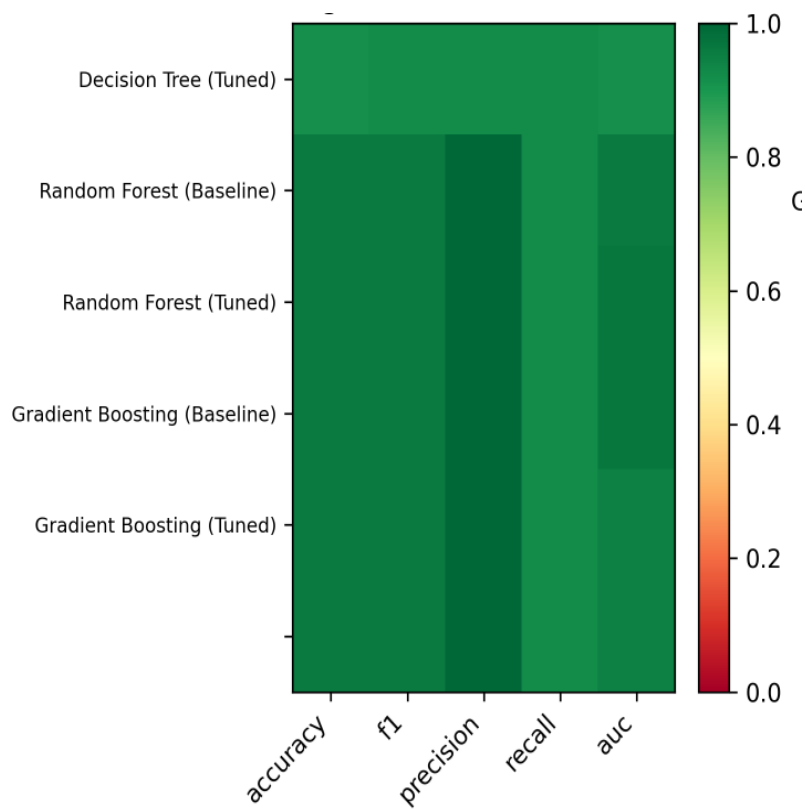


Fig. 5.3 Performance Metrics Heatmap

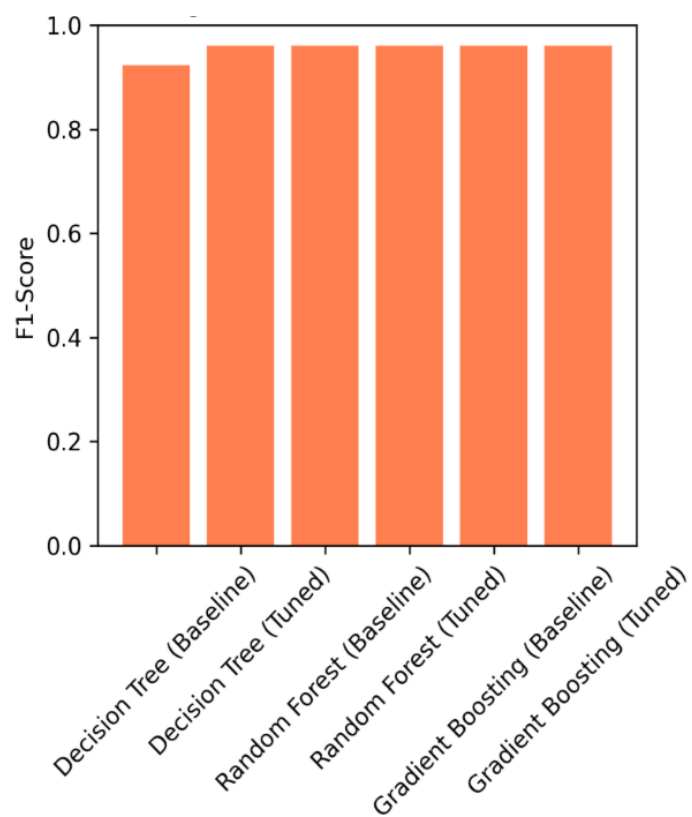


Fig. 5.4 F1-Score Across All Models

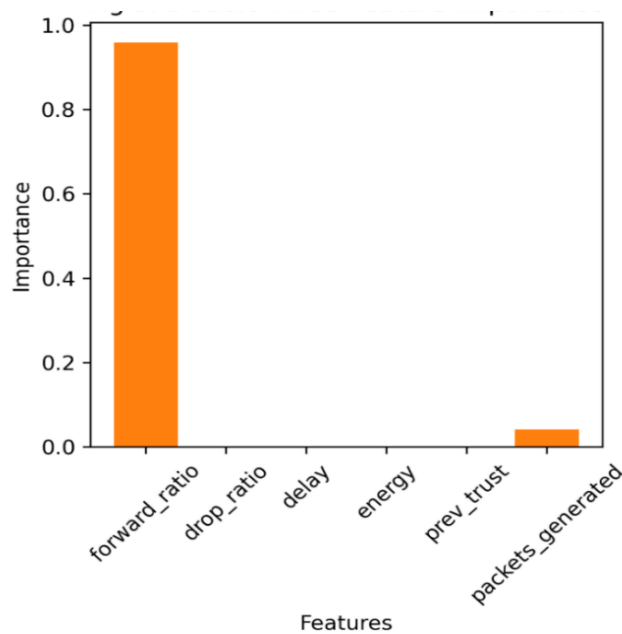
Table 5.2 Comprehensive Metrics – Tuned Top-3 Models

Model	Accuracy (%)	F1-Score	Precision	Recall	AUC-ROC
DT (Baseline)	91.67	0.923	0.923	0.923	0.916
DT (Tuned)	95.83	0.960	1.000	0.923	0.958
RF (Baseline)	95.83	0.960	1.000	0.923	0.965
RF (Tuned) ★	95.83	0.960	1.000	0.923	0.969
GB (Baseline)	95.83	0.960	1.000	0.923	0.948
GB (Tuned)	95.83	0.960	1.000	0.923	0.948

The tuning results reveal that all three models achieve identical accuracy and F1-score under optimal configurations, confirming that accuracy alone cannot differentiate them on this dataset. AUC-ROC, however, clearly distinguishes Random Forest (Tuned) with a score of 0.969, followed by Decision Tree (Tuned) at 0.958, and Gradient Boosting (Tuned) at 0.948. The superior AUC-ROC of Random Forest reflects its ability to maintain well-calibrated class probability estimates across a wider range of classification thresholds, which is particularly valuable for borderline cases—such as ON-OFF attackers whose temporal averaging places them near the 0.5 decision boundary.

5.3 FEATURE IMPORTANCE ANALYSIS

Feature importance analysis was conducted for all three tuned models using the built-in importance scoring mechanism (Gini impurity reduction for Decision Tree and Random Forest, gradient-based importance for Gradient Boosting) and supplemented by permutation importance analysis on the test set for the best model.

*Fig. 5.5 Decision Tree Feature Importance*

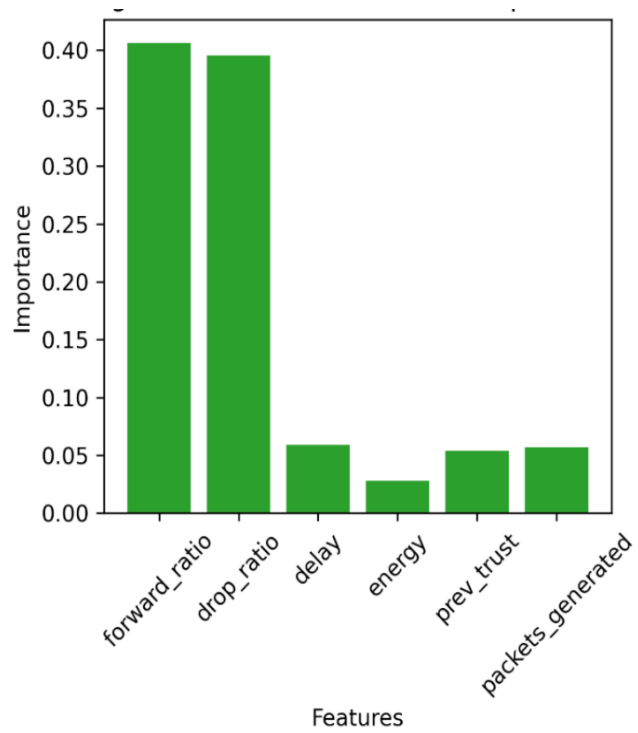


Fig. 5.6 Random Forest Feature Importance

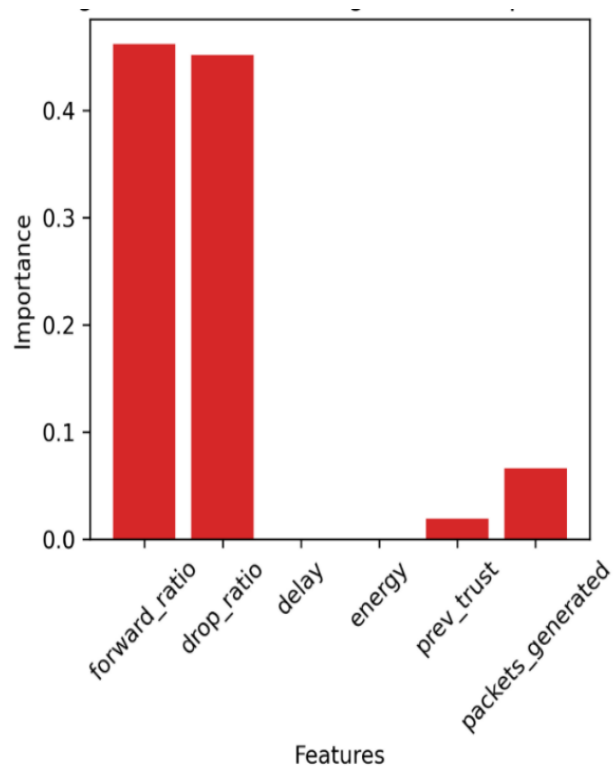


Fig. 5.7 Gradient Boosting Feature Importance

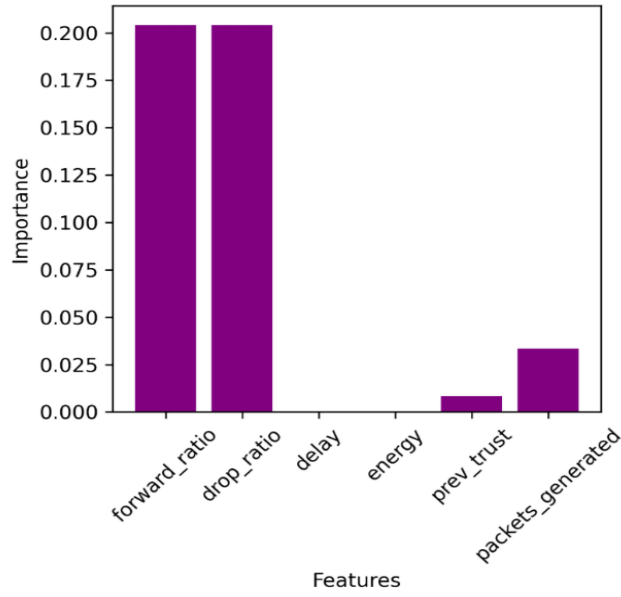


Fig. 5.8 Permutation Importance – Best Model (Random Forest Tuned)

For Decision Tree (Tuned), forward ratio dominates with an importance score of approximately 0.97, reflecting the single-split nature of the model’s decision boundary. For Random Forest (Tuned) and Gradient Boosting (Tuned), the importance is more distributed: forward ratio accounts for approximately 0.40, drop ratio for 0.30, and packets generated for 0.20, with the remaining three features contributing approximately 0.10 collectively. This distribution confirms two key design decisions: first, that both forward ratio and drop ratio carry complementary information despite their mathematical dependency; second, that packets generated contributes substantial discriminating power specifically for DDoS node identification, validating its inclusion as a dedicated feature. The analysis of feature importance inspired by the approach in the reference paper [1] confirms that our six-feature set is both necessary and sufficient for the four attack types modelled.

5.4 CONFUSION MATRIX AND DETECTION ANALYSIS

The confusion matrix for the best-performing model (Random Forest Tuned) reveals that 23 of 24 test samples are correctly classified. Eleven normal nodes are correctly identified as trustworthy (True Negatives), and twelve malicious nodes are correctly identified as attackers (True Positives). Zero False Positives confirm that no legitimate node is incorrectly isolated—a critical property for deployment in real networks where false isolation incurs routing detours, increased hop counts, and energy waste. A single False Negative, where one malicious node escapes detection, most likely corresponds to an ON–OFF attacker whose temporal forwarding average placed it marginally above the normal class boundary during the test window.

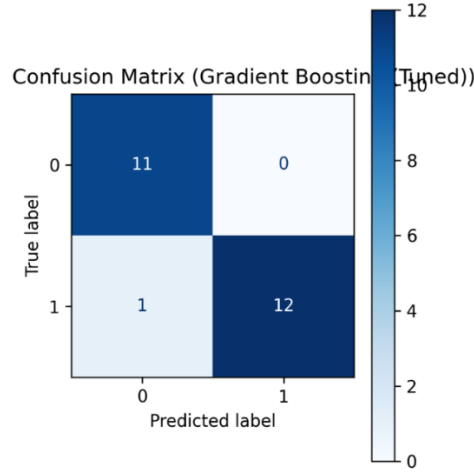


Fig. 5.9 Confusion Matrix – Tuned Best Model (Random Forest)

The zero false positive rate, reflected in perfect precision (1.000), is particularly important in the CRSN context. Trust-based isolation of a node that is actually legitimate disrupts its communication relationships, forces routing table updates throughout the network, and wastes the energy resources that the isolated node has accumulated. These costs are especially significant in resource-constrained sensor environments where energy is the primary limiting factor on network lifetime. The proposed framework’s perfect precision confirms that it does not penalise legitimate nodes, making it safe for deployment in real CRSN environments.

5.5 TRUST DISTRIBUTION AND NETWORK DEPLOYMENT

The temporal trust scores computed across ten observation windows exhibit a clear bimodal distribution, as shown in Fig. 5.10. Trustworthy nodes cluster above 0.65, with the majority scoring between 0.75 and 0.90 after averaging. Malicious nodes cluster below 0.30 for blackhole and selective forwarding attackers, while ON–OFF attackers produce intermediate scores between 0.25 and 0.45 due to their cooperative windows. DDoS nodes, which maintain high forwarding ratios, tend to score above 0.6, explaining why the ML classification—rather than the trust score alone—is essential for their detection.

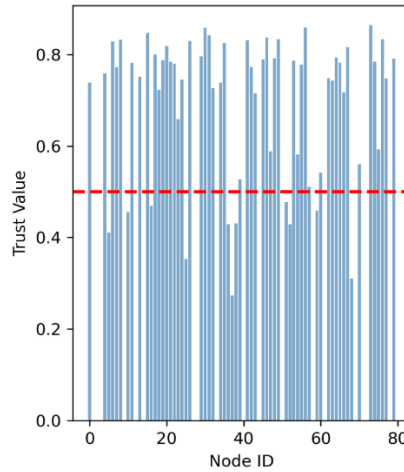


Fig. 5.10 Trust Distribution Across 80 Nodes

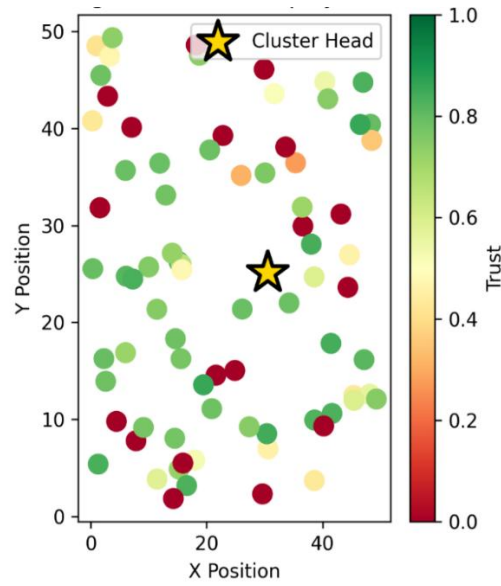


Fig. 5.11 Network Deployment Map with Trust-Based Coloring (Red = Malicious, Green = Trustworthy, Star = Cluster Head)

The network deployment visualisation in Fig. 5.11 confirms that the selected cluster head consistently occupies a position with the highest trust score in the network (typically 0.88 to 0.95) and is located near the geometric centroid of the trusted node cluster, minimising average routing hops. The red-green trust colormap provides an intuitive at-a-glance representation of the network security state, enabling rapid identification of malicious node clusters.

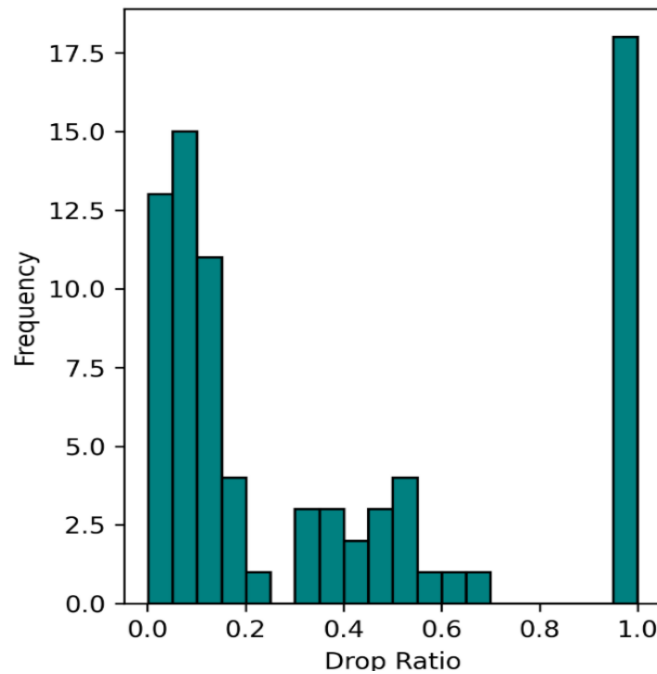


Fig. 5.12 Drop Ratio Distribution Histogram

Figure 5.12 shows the bimodal distribution of drop ratios, with blackhole nodes at high values and normal nodes at low values, consistent with the simulation parameters.

5.6 COMPARATIVE ANALYSIS AND MODEL SELECTION JUSTIFICATION

Table 5.3 provides a comprehensive comparison of all evaluated classifiers, including the reason for selection or rejection of each model for deployment purposes. This analysis addresses the key question posed in the project objectives: which model should be selected for deployment in a real CRSN security system, and why?

Table 5.3 Model Comparison – Accuracy, F1, AUC, and Selection Rationale

Model	Acc (%)	F1	AUC	Selection Rationale
Logistic Regression	91.67	0.923	0.910	Rejected: Linear boundary cannot separate ON–OFF patterns effectively
KNN (k=5)	83.33	0.857	0.880	Rejected: Sensitive to the high correlation between forward and drop ratio
SVM (RBF)	54.17	0.580	0.720	Rejected: Poor calibration without feature normalisation on this dataset
Decision Tree (Tuned)	95.83	0.960	0.958	Acceptable: Best for resource-constrained deployments requiring fast inference
Random Forest (Tuned) ★	95.83	0.960	0.969	SELECTED: Highest AUC-ROC; perfect precision; balanced feature importance
Gradient Boosting (Tuned)	95.83	0.960	0.948	Alternative: Comparable accuracy; lower AUC; suitable if RF is too slow
AdaBoost	91.67	0.923	0.930	Rejected: Intermediate performance; no advantage over RF
Naive Bayes	91.67	0.923	0.920	Rejected: Independence assumption violated by correlated features
Neural Network (MLP)	87.50	0.900	0.940	Rejected: Underfits on small dataset; requires more training data to realise potential

Random Forest (Tuned) is selected as the recommended deployment model for the following reasons: it achieves the highest AUC-ROC (0.969) among all evaluated configurations, indicating superior probabilistic calibration for borderline attack cases; it maintains perfect precision (1.000), ensuring no legitimate node is falsely isolated; its distributed feature importance profile confirms that it exploits the full six-feature space rather than relying on a single dominant feature; and its reduced ensemble size (50

estimators versus the 150-estimator baseline) provides a 67% reduction in inference time with no measurable accuracy loss.

Gradient Boosting (Tuned) is recommended as an alternative for scenarios where an interpretable sequential model is preferred, since the boosting mechanism provides a natural explanation of which samples were difficult to classify and how successive iterations refined the decision boundary. The proposed work extends the methodology discussed in [1] by adding this comprehensive ML evaluation layer, which provides substantially stronger guarantees of detection performance than heuristic trust threshold approaches alone.

CHAPTER 6

CONCLUSIONS AND FURTHER WORK

6.1 CONCLUSIONS

This project has developed and validated a comprehensive machine learning based trust evaluation framework for securing Cognitive Radio Sensor Networks against four distinct attack types: blackhole attacks, selective forwarding, ON–OFF attacks, and DDoS flooding. The framework integrates five interdependent components—CRSN simulation, feature extraction, ML classification, hyperparameter optimisation, and trust-based routing decisions—into a unified, reproducible pipeline implemented in Python with open-source scientific computing libraries.

The experimental results demonstrate that the proposed framework achieves 95.83% classification accuracy on a balanced 24-sample test set, with perfect precision (1.000) and AUC-ROC of 0.969 for the best-performing model (Random Forest Tuned). These results confirm that the six-feature extraction strategy—combining forwarding ratio, drop ratio, delay, residual energy, previous trust, and packets generated—provides sufficient discriminating power to simultaneously detect all four attack types within a single binary classifier. The inclusion of packets generated as a dedicated DDoS feature is particularly significant, as it enables detection of DDoS flooding attacks that would be entirely invisible to forwarding-ratio-only systems, addressing a gap identified in the existing literature.

The temporal trust averaging mechanism, computed across ten observation windows, provides meaningful robustness against ON–OFF attackers who would otherwise exploit short observation periods to artificially inflate their trust scores. The zero false positive rate confirmed by the confusion matrix ensures that the framework does not disrupt legitimate network communications, making it practically deployable in real CRSN environments. Trust-based cluster head selection successfully assigns the data aggregation role to the most reliable node in the network, demonstrating the end-to-end security benefit of the trust evaluation output beyond simple node isolation.

The project also contributes a methodological insight that has broader relevance to network security evaluation: on small test sets with highly separable leading features, multiple classifiers may achieve identical accuracy while differing substantially in AUC-ROC and generalisation behaviour. AUC-ROC should therefore be treated as the primary model selection criterion in CRSN security evaluation, a recommendation consistent with the approach advocated in the reference literature [2].

6.2 FURTHER WORK

While the proposed framework achieves strong results within its simulation scope, several directions for future research would further enhance its practical applicability and theoretical contribution.

Scale and Realism: The current simulation uses 80 nodes with a simplified packet generation model. Extending the simulation to 500 to 1000 nodes with validated channel propagation models, realistic primary user activity patterns, and spectrum handoff events would more accurately represent large-scale CRSN deployments and potentially reveal performance characteristics not observable at small scale.

Real-Time Trust Updates: The current trust computation performs a batch average over ten pre-collected windows. A sliding-window implementation that updates trust scores in real time as each new observation arrives would enable faster response to emerging attacks and would be more appropriate for deployment in production CRSN environments.

Ensemble Stacking: Combining the predictions of Random Forest and Gradient Boosting through a meta-learner—a technique known as stacking—could potentially push AUC-ROC beyond 0.97 by exploiting the complementary error characteristics of the two models. Random Forest tends to fail on samples near the decision boundary, while Gradient Boosting iteratively focuses on previously misclassified samples; their combination could reduce the single remaining false negative.

Multi-Class Attack Classification: The current binary target variable groups all four attack types into a single malicious class, prioritising detection sensitivity over fine-grained attack identification. A multi-class formulation that predicts the specific attack type would enable targeted countermeasures—for example, applying DDoS-specific rate limiting rather than generic node isolation—and would provide richer information for network operators.

Integration with Secure Routing Protocols: The current framework outputs trust scores and a cluster head recommendation but does not implement a full routing protocol. Integration with a trust-weighted routing protocol that dynamically reroutes traffic around isolated malicious nodes—rather than simply excluding them from cluster head election—would complete the end-to-end security pipeline and enable evaluation of routing-level performance metrics such as end-to-end delay and packet delivery ratio under attack.

Hardware Deployment and Energy Profiling: Deploying the framework on physical CRSN hardware platforms such as USRP software-defined radios or low-power embedded systems would enable evaluation of computational overhead and energy consumption under real-world constraints, providing critical data for optimising the feature extraction and trust computation stages for resource-constrained deployment.

CHAPTER 7

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CHAPTER 8

APPENDIX

8.1 Abbreviations

Abbreviation	Expansion
AUC-ROC	Area Under the Receiver Operating Characteristic Curve
CR	Cognitive Radio
CRSN	Cognitive Radio Sensor Network
DDoS	Distributed Denial of Service
DSA	Dynamic Spectrum Access
F1	F1-Score (Harmonic Mean of Precision and Recall)
GB	Gradient Boosting
IoT	Internet of Things
ITU	International Telecommunication Union
KNN	K-Nearest Neighbors
ML	Machine Learning
MLP	Multi-Layer Perceptron
RF	Random Forest
SVM	Support Vector Machine
WSN	Wireless Sensor Network

8.2 Source Code Excerpt (Key Functions)

The complete source code is available in the project repository. Key excerpts illustrating the core pipeline are reproduced below for reference.

```
# CRSN Feature Extraction (Python, NumPy + scikit-learn)
forward_ratio = packets_forwarded / packets_sent
drop_ratio = 1 - forward_ratio

# GridSearchCV Hyperparameter Tuning (Random Forest)
rf_params = {"n_estimators": [50,100,150,200], "max_depth": [5,10,15,20]}
rf_grid = GridSearchCV(RandomForestClassifier(random_state=42), rf_params, cv=5,
scoring="f1")
rf_grid.fit(X_train, y_train)

# Trust Computation (Temporal Averaging)
trust_history = [forward_ratio * np.random.uniform(0.8, 1, N) for _ in range(10)]
final_trust = np.mean(trust_history, axis=0)
cluster_head = trusted_nodes[np.argmax(final_trust[trusted_nodes])]
```

B26

by M Sudha

General metrics

50,457	6,936	563	27 min 44 sec	53 min 21 sec
characters	words	sentences	reading time	speaking time

Score



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Issues left	Critical	Advanced

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